

LILIAN ALHALABI

Data Scientist | AI & Machine Learning

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PROFESSIONAL SUMMARY

Data Science graduate (First Class Honours, Artificial Intelligence Track) from Saudi Electronic University with hands-on experience in machine learning, NLP, and data analytics. Completed a Data Engineer Trainee program at IT-Ranks Technology, gaining practical experience with SQL, Docker, Debezium/Kafka change-data-capture testing, and BI dashboarding tools. Led a senior capstone project on Arabic sentiment analysis (Python, NLP, TensorFlow, PyTorch), selected for presentation at the Geneva graduation-projects exhibition in 2027. Interested in developing ML and AI solutions, with particular interest in NLP, Generative AI, and applied AI.

SKILLS

Programming: Python, SQL

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, NLP, Machine Learning, Statistical Analysis

Databases: MySQL, SQL

Visualization & BI: Power BI, Apache Superset, Metabase, Cube.js, Excel

Data Engineering: ETL/ELT Concepts, Data Generation, Docker, Trino, Debezium, Apache Kafka, Apache Airflow (Fundamentals)

Additional Exposure: Apache Iceberg, dbt, ClickHouse

EXPERIENCE

Data Engineer Trainee — IT-Ranks Technology, Riyadh, Saudi Arabia

Jun 2026 – Aug 2026

- Wrote Python and SQL scripts for data engineering tasks, including querying, manipulating, and validating data across multiple databases; built and used a custom data generator to produce sample datasets for pipeline testing.
- Created and tested a Debezium connector in a local environment, observing change-data-capture events flow through Apache Kafka.
- Queried and joined data across multiple sources using Trino at a basic level; worked hands-on with Docker in local environments, troubleshooting container, configuration, and runtime issues.
- Independently studied Apache Airflow fundamentals, explored existing DAGs, and presented key workflow concepts to the data team; gained introductory exposure to Apache Iceberg, dbt, and ClickHouse.

PROJECTS

Sentiment Analysis Engine — Senior Capstone

2026

Python, NLP, TensorFlow, PyTorch, Scikit-learn, MARBERT, Streamlit — Selected for presentation at the Geneva graduation-projects exhibition (2027).

- Led a student team to develop an Arabic-language sentiment analysis engine for e-commerce customer reviews.
- Preprocessed and classified Arabic review text using NLP techniques and the MARBERT language model, comparing approaches across TensorFlow, PyTorch, and Scikit-learn.
- Built and demonstrated a Streamlit interface using Google Colab and Cloudflare Tunnel; received strong feedback from the evaluation committee.

Analytics Solution — Confidential Project (Data Engineer Trainee)

2026

MySQL, Cube.js, Apache Superset, Metabase, Docker — Confidential project; business details withheld.

- Developed a containerized analytics solution in a local training environment: generated and validated sample data via SQL, then used Docker and Cube.js as a semantic layer over a relational database.
- Built dashboards in Apache Superset and Metabase on top of Cube.js, defining metrics and dimensions and validating consistency between source data and dashboard outputs.

Heart Disease Prediction — Machine Learning

2026

Python, Scikit-learn, Pandas — Built and evaluated classification models to predict heart disease risk from clinical data; performed preprocessing, feature selection, and algorithm comparison.

EDUCATION

Bachelor of Data Science | Artificial Intelligence Track — Saudi Electronic University, Riyadh

2026

GPA: 3.88 / 4.0 | First Class Honours

Apple Foundation Program, Apple Developer Academy — Riyadh, Saudi Arabia

2025

Junior iOS Developer — Collaborated within a team to design and build a functional mobile application.

CERTIFICATIONS

- IBM Data Science Professional Certificate — IBM (2024)
- Elements of AI — University of Helsinki & Google (2025)
- SQL for Data Science — University of California, Davis (2026)
- Data Analysis and Visualization with Power BI — Microsoft (2026)

Languages: Arabic (Native) | English (Fluent)